UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/570,125	01/06/2022	Yu Fang	SONA-0006-U01-C04	6430

87084 7590 07/20/2023

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
NGUYEN, ANGELA	

ART UNIT	PAPER NUMBER
2442	

NOTIFICATION DATE	DELIVERY MODE
07/20/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/570,125	Applicant(s) Fang et al.	
	Examiner ANGELA NGUYEN	Art Unit 2442	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-26 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-26 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 1, 6, 7, 13, 16, 25 and 26 is/are rejected under 35 U.S.C. 103 as being unpatentable over Mo et al. (U.S. 2021/0144068), hereinafter Mo in view of Olson et al (U.S. 2019/0129778), hereinafter Olson.

With respect to Claim 1, Mo teaches a system comprising:

a vehicle having a first network and a second network (¶ 0056, "In addition, a network control instruction is transmitted from an active network and a standby network to all slave nodes");

and a communication device interposed between the first network and the second network (figure 4, active or standby master node) and structured to:

generate network status data by monitoring communications of one or more devices on the first and/or the second networks (¶ 0147, "If no PDO packet transmitted by the first node is received on the standby network within the preset second heartbeat period corresponding to the first node, transmit current fault messages of the first CAN channel and the second CAN

channel of the first node to the operation monitoring node, and display the current fault messages to the operator as a prompt of a current fault to be fixed.")

and transmit the network status data to a sensor device external to the vehicle ¶ 0147, "If no PDO packet transmitted by the first node is received on the standby network within the preset second heartbeat period corresponding to the first node, transmit current fault messages of the first CAN channel and the second CAN channel of the first node to the operation monitoring node, and display the current fault messages to the operator as a prompt of a current fault to be fixed.")

Mo fails to explicitly teach a second network of a different type than the first network.

Olson teaches a second network of a different type than the first network (¶ 0009, "It is noted, however, that other networks and other redundant buses employing other bus communication protocols may be used, in accordance with further implementations.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the invention of Olson as having a redundant bus of a different type allows for multiple points of failure and reduced cost as some devices will come with multiple types of ports to allow interconnection and an existing port can be used.

With respect to Claim 6, Mo in view of Olson teaches the system of claim 1, Mo teaches wherein the communication device is further structured to determine an event value based at least in part on the monitored communications of the one or more devices on the first and/or the second networks and including the event value in the network status (¶ 0076, "The count to

255 of the error counter is merely an example. Depending on different application requirements, when the count of the transmission error counter or the receipt error counter of the active master node adds up to any preset value that meets requirements.")

With respect to Claim 7, Mo in view of Olson teaches the system of claim 6, wherein the event value is based at least in part on an error condition (¶ 0074, "When a bus of the active network of the active master node is faulty (for example, voltage of a CAN bus is abnormal, bus-off caused by too many frame errors, and the like.")

With respect to Claim 13, Mo in view of Olson teaches the system of claim 1, wherein the first network and/or the second network is controller area network (CAN) based (¶ 0033, "A train network node and a CANopen-based train network node monitoring method according to embodiments of the present disclosure are described below with reference to accompanying drawings.")

With respect to Claims 16, 25, and 26, claims 16, 25 and 26 are the apparatus that corresponds to the system of Claims 1, 6 and 17 and are rejected accordingly

Claim(s) 2-4, 14, 15, 17 and 18 is/are rejected under 35 U.S.C. 103 as being unpatentable over Mo et al. (U.S. 2021/0144068), hereinafter Mo in view of Olson et al (U.S. 2019/0129778), hereinafter Olson, in view of Zinner et al. (U.S. 2015/0228130), hereinafter Zinner.

With respect to Claim 2. Mo in view of Olson teaches the system of claim 1, wherein: the communication device comprises a plurality of ports on the first network; and the communication device is further structured to configure at least a first port of the plurality to mirror at least a second port of the plurality

Zinner teaches the communication device comprises a plurality of ports on the first network; and the communication device is further structured to configure at least a first port of the plurality to mirror at least a second port of the plurality (¶ 0066, "Port mirroring makes it possible to mirror the network traffic from one or more ports (inputs) of a switch to another port (which is also called mirror port). With active port mirroring, a data rate would theoretically be 90 Mbit/s+70 Mbit/s=160 Mbit/s and would therefore be 60 Mbit/s above the data rate which can actually be transported. Switch 18a will accordingly reject data packets and not all data packets will arrive at the diagnostic device 12.".)

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the system of Zinner as Ethernet is a common networking system as the other network and it would be obvious to incorporate it and interwork between it as it increases the industrial applicability.

With respect to Claim 3, Mo in view of Olson in view of Zinner teaches the system of claim 2, wherein: the communication device comprises a plurality of ports on the first network; and the communication device is further structured to: assign a first device, of a plurality of devices, on the second network to a port of the plurality; and transmit the network status data to the sensor device on the first network via at least one port of the plurality

Zinner teaches the communication device comprises a plurality of ports on the first network; and the communication device is further structured to: assign a first device, of a plurality of devices, on the second network to a port of the plurality; and transmit the network status data to the sensor device on the first network via at least one port of the plurality (¶ 0066, "Port mirroring makes it possible to mirror the network traffic from one or more ports (inputs) of a switch to another port (which is also called mirror port). With active port mirroring, a data rate would theoretically be 90 Mbit/s+70 Mbit/s=160 Mbit/s and would therefore be 60 Mbit/s above the data rate which can actually be transported. Switch 18a will accordingly reject data packets and not all data packets will arrive at the diagnostic device 12.".)

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the system of Zinner as Ethernet is a common networking system as the other network and it would be obvious to incorporate it and interwork between it as it increases the industrial applicability.

With respect to Claim 4, Mo in view of Olson in view of Jeon teaches the system of claim 3, wherein the second network is arranged in a bus topology and the first network is arranged in a star topology

Zinner teaches the second network is arranged in a bus topology and the first network is arranged in a star topology (¶ 0018, "In the case of relatively large volumes of data for example, vehicle access via Ethernet may prove to be very advantageous as new diagnostic access. Introduction of diagnostic access can be advanced even more quickly with the introduction of

standardized IP diagnostic interfaces, as specified in ISO 13400, using Ethernet as the physical layer.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the system of Zinner as Ethernet is a common networking system as the other network and it would be obvious to incorporate it and interwork between it as it increases the industrial applicability.

With respect to Claim 14. Mo in view of Olson teaches the system of claim 1, however, fails to teach wherein the first network and/or the second network is Ethernet based.

Zinner teaches the first network and/or the second network is Ethernet based (¶ 0018, "In the case of relatively large volumes of data for example, vehicle access via Ethernet may prove to be very advantageous as new diagnostic access. Introduction of diagnostic access can be advanced even more quickly with the introduction of standardized IP diagnostic interfaces, as specified in ISO 13400, using Ethernet as the physical layer.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the system of Zinner as Ethernet is a common networking system as the other network and it would be obvious to incorporate it and interwork between it as it increases the industrial applicability.

With respect to Claim 15, Mo in view of Olson teaches the system of claim 1, Mo teaches wherein the first network or the second network is controller area network (CAN) based (¶ 0033, "A train network node and a CANopen-based train network node monitoring

method according to embodiments of the present disclosure are described below with reference to accompanying drawings.")

Mo in view of Olson fails to teach the other of the first network and the second network is Ethernet based.

Zinner teaches the other of the first network and the second network is Ethernet based (¶ 0018, "In the case of relatively large volumes of data for example, vehicle access via Ethernet may prove to be very advantageous as new diagnostic access. Introduction of diagnostic access can be advanced even more quickly with the introduction of standardized IP diagnostic interfaces, as specified in ISO 13400, using Ethernet as the physical layer.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the system of Zinner as Ethernet is a common networking system as the other network and it would be obvious to incorporate it and interwork between it as it increases the industrial applicability.

With respect to Claims 17 and 18, Claims 17 and 18 are the apparatus that corresponds to the system of Claims 2 and 3 and are rejected accordingly.

Claim(s) 5, 8-12 and 19-24 is/are rejected under 35 U.S.C. 103 as being unpatentable over Mo et al. (U.S. 2021/0144068), hereinafter Mo in view of Olson et al (U.S. 2019/0129778), hereinafter Olson, in view of Flauaus et al. (U.S. 2005/0108444), hereinafter Flauaus.

With respect to Claim 5, Mo in view of Olson teaches the system of claim 1, wherein the communication device is further structured to determine a statistical value based at least in

part on the monitored communications of the one or more devices on the first and/or the second networks and including the statistical value in the network status

Flauaus teaches the communication device is further structured to determine a statistical value based at least in part on the monitored communications of the one or more devices on the first and/or the second networks and including the statistical value in the network status (¶ 0062, “By providing the history data in this manner, multiple types of management platforms are able to calculate a change in congestion management statistics independently and simultaneously without impacting the switch's management period. Thus if management platform "A" wanted to look at the change in congestion statistics every 10 minutes and management platform "B" wanted to compare the congestion statistics changes every minute, each management application may do so by refreshing their congestion statistics at their fixed durations (10 minutes and 1 minute respectively) and comparing the latest sample with the previous retained statistics.”)

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo and Olson with the invention of Flauhaus to allow for better administration and monitoring of networked devices in the system of Mo.

With respect to Claim 8, Mo in view of Olson teaches the system of claim 1, wherein the communication device is further structured to: modify the network status data.

Flauaus teaches the communication device is further structured to: modify the network status data (¶ 0062, “By providing the history data in this manner, multiple types of

management platforms are able to calculate a change in congestion management statistics independently and simultaneously without impacting the switch's management period. Thus if management platform "A" wanted to look at the change in congestion statistics every 10 minutes and management platform "B" wanted to compare the congestion statistics changes every minute, each management application may do so by refreshing their congestion statistics at their fixed durations (10 minutes and 1 minute respectively) and comparing the latest sample with the previous retained statistics.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo and Olson with the invention of Flauhaus to allow for better administration and monitoring of networked devices in the system of Mo.

With respect to Claim 9, Mo in view of Olson in view of Flauhaus teaches the system of claim 8, wherein the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to at least one device, on the first or second network, identified by the data selection command value

Flauaus teaches the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to at least one device, on the first or second network, identified by the data selection command value (¶ 0064, "Box 610 represents a "well behaved device" in which a port has no unusual traffic patterns and utilization is not high. Box 614

illustrates an F_Port that is identified as congested in the RX direction but since link utilization is low, the module 230 determines that the cause is a busy device elsewhere and the congestion type backpressure (which is generated by the port in the RX direction).")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo and Olson with the invention of Flauhaus to allow for better administration and monitoring of networked devices in the system of Mo.

With respect to Claim 10, Mo in view of Olson in view of Flauhaus teaches the system of claim 8, wherein the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to one or more systems of the vehicle identified by the data selection command value.

Flauaus teaches the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to one or more systems of the vehicle identified by the data selection command value. (¶ 0064, "Box 610 represents a "well behaved device" in which a port has no unusual traffic patterns and utilization is not high. Box 614 illustrates an F_Port that is identified as congested in the RX direction but since link utilization is low, the module 230 determines that the cause is a busy device elsewhere and the congestion type backpressure (which is generated by the port in the RX direction).")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo and Olson with the invention of Flauhaus to allow for better administration and monitoring of networked devices in the system of Mo.

With respect to Claim 11, Mo in view of Olson in view of Flauhaus teaches the system of claim 8, wherein the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to at least one protocol identified by the data selection command value

Olson teaches the communication device is further structured to: selectively modify the network status data in response to a data selection command value; and include data in the network status data that corresponds to at least one protocol identified by the data selection command value (¶ 0009, "It is noted, however, that other networks and other redundant buses employing other bus communication protocols may be used, in accordance with further implementations.")

Therefore, it would have been obvious to one of ordinary skill in the art prior to the filing of the application to combine the invention of Mo with the invention of Olson as having a redundant bus of a different type allows for multiple points of failure and reduced cost as some devices will come with multiple types of ports to allow interconnection and an existing port can be used.

With respect to Claim 12, Mo in view of Olson in view of Flauhaus teaches the system of claim 11, wherein the at least one protocol comprises a controller area network (CAN) or an on-board diagnostic (OBD) (¶ 0033, "A train network node and a CANopen-based train network node monitoring method according to embodiments of the present disclosure are described below with reference to accompanying drawings.")

With respect to Claims 19-24, claims 19-24 are the apparatus that corresponds to the system of Claims 5, 8-12 and is rejected accordingly.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANGELA NGUYEN whose telephone number is (571)270-5660. The examiner can normally be reached Monday - Friday 9AM - 5PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William Trost can be reached on 571-272-7872. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available

to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ANGELA NGUYEN/
Primary Examiner, Art Unit 2442